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Tutorial 2 (Classes on 12, 13, 19 & 20th of August 2025): Solutions

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1 Fill in the blanks

1. The most common processor instruction (which occurs 90 % of the time) is `data flow` (data flow / addition / Ex-OR / multiplication).
2. The output of two or more gates should not be connected together, `because of the possibility of short circuit due to difference in the output voltage`.
3. Tri-state gates are solutions `to short circuit due to the possibility of differences in the output voltage (of gates) when outputs of more than one gates are connected`.
4. A single bit data bus interfacing the N (N>3) flip-flops is created by `connecting the inputs and outputs of FF` while the control signals involved are `RD and WR`. To transmit data from one FF to another, the procedure is `(a) only the source FF's RD is ON and upon the clock pulse hitting the FF means data is READ into the data bus (b) in the second part, the destination FF is selected by asserting the WR control input and is written into when the clock pulse hits`
5. The purpose of buffer registers connected to data bus `is to allow data to be held temporarily, ensuring that the sender and receiver communicate in the same clock interval (but in disjoint sub-intervals) making possible reliable data transfer. This prevents data loss or corruption or interference among multiple devices sharing/ accessing the data bus.`
6. In N (=16) 8-bit registers connected through 8-bit data bus, the width of address bus is `4 bits,` width of data bus is `8 bits.`
7. Security of the processor is more in `Harvard` (Harvard / von Neumann) architecture and the reason(s) is (are): `program is stored in ROM and is read-only. (Data are stored in separate memory - RAM). Data & program memories have dedicated (independent) buses to access them). Because of this, any eavsdropper's attempts to change the program is not possible (malaware).`
8. AVR microcontroller complies to `Harvard` (Harvard / von Neumann) architecture. Hence, the program memory and data are stored in the `different` (same / different) memory location.
9. The purpose of the functions lines in ALU is to `specify the current ALU operation`. Function lines are generated by `Control Unit (CU)`
10. Exactly 1 KB means `1024` bytes.
11. The binary and decimal equivalent of hex numbers 0x 2B9 is $(001010111001)_2$ and $(697)_{10}$
12. In Load / Store architecture (LSA) apart from LD and ST instructions other instructions `can't` (can also / can't) access the memory.
13. In LSA, `none of` (either one or both or none of) the operands `can` (can / can't) be from main memory.
14. In register-memory architecture (RMA), `either one or both` (either one or both or none of) the operands `can` (can / can't) be from main memory.

15. Which of the following is (are) volatile? (a) SRAM, (b) EEPROM (c) DRAM (d) NV-RAM

    `SRAM and DRAM`

16. Of the instructions (a) ADD R0, R1 (b) MOV R15, R16 (c) LDI R0, K, (b) MOV R15, R16 (c) LDI R0, K, are data transfer operation(s) ADD R0, R1 is ALU arithmetic operation(s).

2 Answer the following

1. Consider the AVR ATmega8 processor, which has the flash memory (program ROM) of 8KB. Given that, in AVR the word size in program ROM is two bytes,

   (a) what is the role of the PC in muP?

   (b) is the PC (a register), accessible to programmer? If it is so, what is it's address in ATmega8? Where does it sit? Within the muP or external SRAM or ROM?

   (c) what is the memory of flash in terms of bytes? What is the maximum number of instructions, that can be accommodated in the given program ROM?

   (d) What is the size of PC (in bits)?

   (e) What is the size of program address bus?

   Solutions:

   (a) `The role of PC is to keep track of next instruction to be executed`

   (b) `No.  NA. It sits in the control unit of CPU and is not visible / accessible by programmer.`

   (c) `The size of PC is 12 bits.`

   (d) `The size of program address bus is 12 bits.`

2. A 512K memory chip has 8 pins for data. Find

   (a) the organization (meaning, how many words versus word length) and

   (b) the number of address pins for this memory chip

   Solutions:

   (a) `The word length is 8 bits.  (Assumed the memory chip is of size 512KB). Then the number of rows is 512 x 1024 = 524288 lines.`

   (b) `The number of address pins` $= log_2(524288) = log_2\left(2^{19}\right) = 19$

3. In a given byte-addressable microcomputer, memory locations 0x 10000 to 0x 9FFFF are available for user programs. The first location (address) is 0x 10000 and the last location (address) is 0x 9FFFF. Calculate,

   (a) the total amount of memory (in bytes) available for user programs

   (b) the word length

   (c) the size of address bus

   (d) the size of data bus

   Solutions:

   (a) `the total amount of memory (in bytes)` $= 9 \times \left(2^{16}\right)$

   (b) `the word length` $= 8$ `bits`

   (c) `the size of address bus` $= ceil\left[log_2\left(9 \times 2^{16}\right)\right]$

   (d) `the size of data bus` $= 8$ `bits`

4. Consider a flip-flop. Assume active high logic. Can the control signal "WR / latch enable" and "RD enable", both high at the same time? Both of them low at the same time?

   Solutions:

   `The control signals ''WR / latch enable'' and ''RD enable'', for a flip-flop cant be asserted at the same time. This could be ensured by having a single line (high for WR and low for RD), but such designs are rare. But, both of them could be low (as in the case of chip disable signal).`

5. How is a single 8-bit register built out of flip-flips? Show the control lines to the register and show how is it connected to RD & WR of the individual flip-flops. Mark the data lines.

   Solutions:

   `Its fairly easy. Please see class notes / presentation files.`

6. Given a set of n $(= 2^m)$ registers (each of (word) length k bits) for some $m = 0, 1, 2, \ldots,$

   (a) What is the memory size in bits? And in bytes when $k = 8$?

   (b) What is the minimum number of address bits necessary to pick an unique register from the above set? Maximum number ..?

   (c) Evolve an addressing scheme which uses minimum bus size. Show the digital circuit derived indicating the TT if any, at all used in the design.

   (d) Give an overall block diagram indicating the various units used in (b) above viz., register bank, address bus, data bus and control lines / bus. Indicate the size of each bus.

   (e) Instead of separate "WR / latch enable" and "RD enable" physical wireline, what if one has a single physical line along with the 'rule', READ for logical 'low' and WR for logical 'high'. What are the pros and cons of such a scheme?

   (f) Explain how the steps involved in instruction MOV Rd, Rs when executed, by using the above block diagram. Here, Rd & Rs are registers of size l bits. Identify the physical path of the digital signal journey.

   (g) Show the timing diagram of execution of the instruction in (d). [Timing diagram has not been taught in the class and hence this is optional].

   `Solutions:`

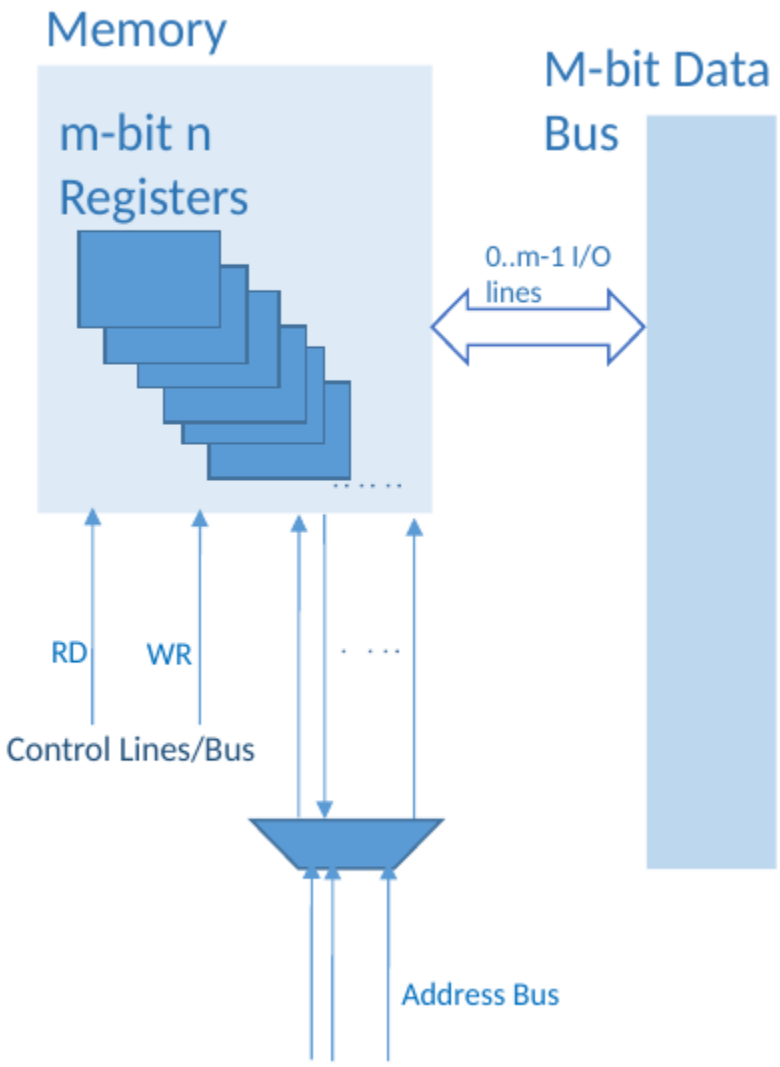
   (a) memory size is kn $(= k \times 2^m)$ bits. When k=8, the memory size is $8n/8$ $(= n = 2^m)$ bytes

   (b) Minimum number of address bits, is m bits. (Maximum number of address bits is ∞).

   (c) Use the m to 2^m decoder, to pick a given register out of n such registers. This means that the decoder needs only m bit input. Corresponding to 2 - 4 decoder,

A_1	A_0		Y_3	Y_2	Y_1	Y_0
0	0		0	0	0	1
0	1		0	0	1	0
1	0		0	1	0	0
1	1		1	0	0	0

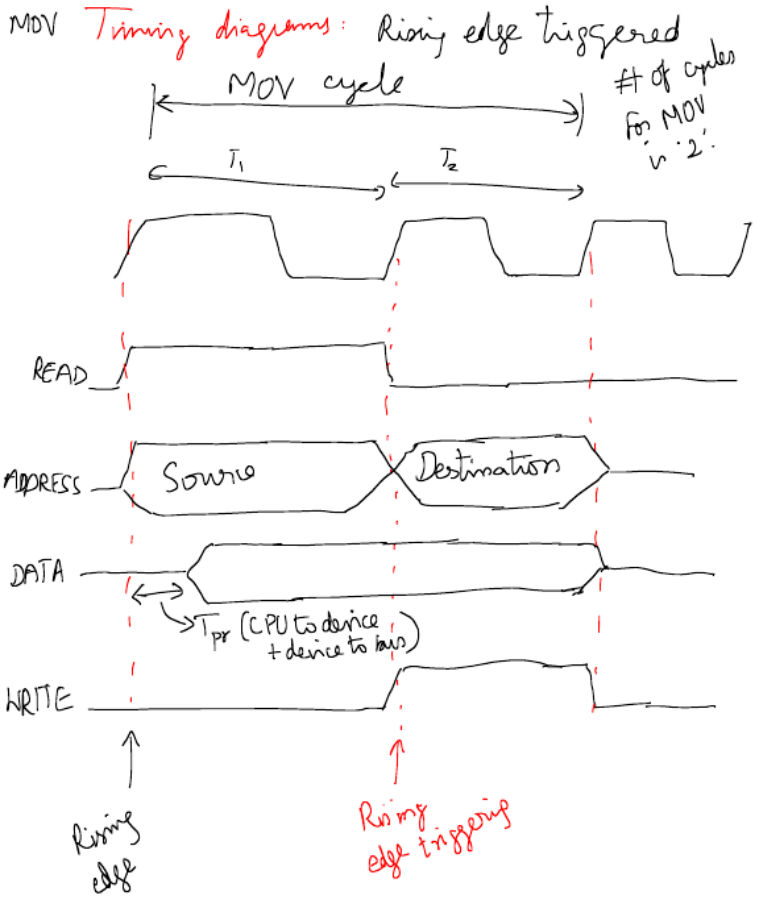
   (d) Again refer week3 presentation file. The size of address bus is $ln_2(n)$ bits. Size of data bus, in figure below, is m=8 bits. Size of control bus is 4 (WR, RD, CS, Clk). (The arrow pointing down in output of decoder is wrong).



(e) We can have such an implementation. The pros of such an implementation is that the number of lines in control bus is reduced by one and thus results in reduced footprint. Another advantage is that both RD and WR can't happen in this approach. Cons is that both of them cant be zero at a time, which is a control input combination required at times. But, one can use CS to 'mute' the memory.

(f) The MOV opcode takes two clock cycles, (1) (a) READ asserted (1)(b) Rs Source ADDRESS placed on address bus (1)(c) Target source device responds after the round trip propagation delay (delay in reaching device + gate delay + propagation delay from device to bus). (2)(a) Rs Source ADDRESS on address bus removed (2)(b) Target destination device Rd ADDRESS placed (2)(c) READ signal deactivated (2)(d) WRITE signal asserted (2)(e) Data written into the destination device (3)(a) Data line is reset (all lines = 0, DATA_BUS cleared) (3)(b) WRITE signal deactivated (Control lines cleared). (3)(c) Destination ADDRESS removed (Address lines cleared). [This has to be marked in the figure showing source R_s and destination registers R_d within register bank, (address, data & control) buses].

(g) Timing diagram is given below and is self-explanatory.



7. Consider a 8-bit processor in which the data bus is connected to internal 8-bit registers whose address range from 0x 0000 till 0x FFFF.

   (a) Show the block diagram as well as circuit diagram of the above system, showing the registers, data bus, control lines and address bus. Indicate quantitatively the various system parameters.

   (b) What is the word length here? What is the size of address bus and data bus?

   (c) Show the hardware level steps (include the **primitive control signal level steps**) to execute the following instructions, given that the register R0 is at address `0x 0000`:

      i. `LDS R0, 0x 0300`

      ii. `MOV R0, R1`

   Solutions:

   (a) `See the wk3 class ppt.`

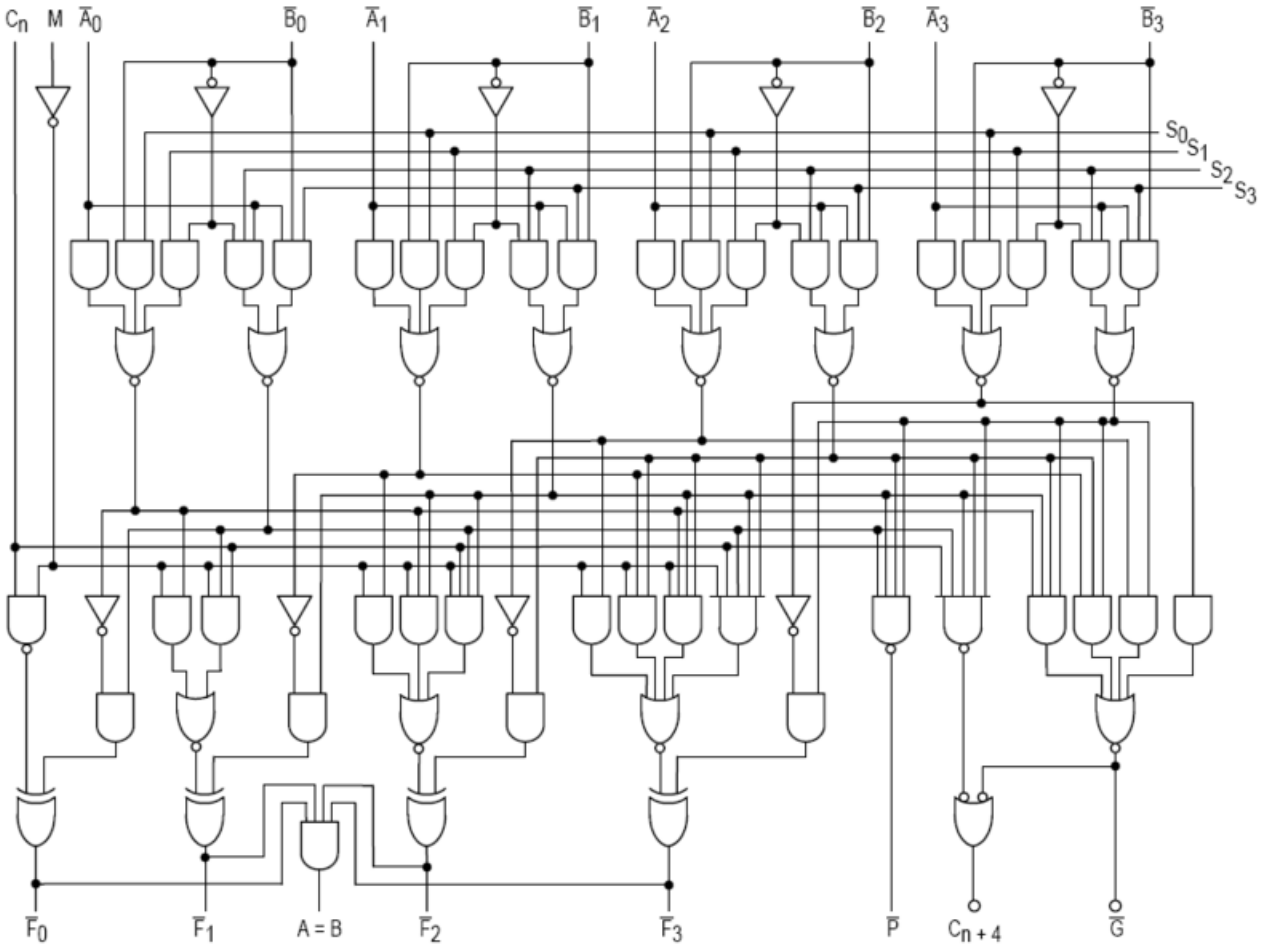
   (b) `8-bits. Size of address bus is '16', and that of data bus is '8'.`

   (c) Control signal level steps

      i. `Consider LDS R0, 0x 0300 This program loads R0 with the value stored in main memory location at the address 0x 0300.`

      ii. `The next instruction MOV R0, R1 copies the value of the register R1 into the register R0`

8. Shown below is the circuit diagram of 1960's famed 4-bit ALU 74181 MSI.



Consider the following table of inputs:

					Active-high logic	Active-high logic
	S_3	S_2	S_1	S_0	Logic $(M = 1)$	Arithmetic $(M = 0)(C_n = 1)$
	0	0	0	1	output F = ?	output F = ?
	0	1	1	0	output F = ?	output F = ?
	1	0	0	1	output F = ?	output F = ?
	1	1	0	0	output F = ?	output F = ?

Assuming active-high logic, enter the value of $F_3F_2F_1F_0$ in terms of $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ for the cases given in the above table. You could use the short hand notation of A for $A_3A_2A_1A_0$ and similarly for B. Give the pathway of the digital signal as it traverses from the input to output.

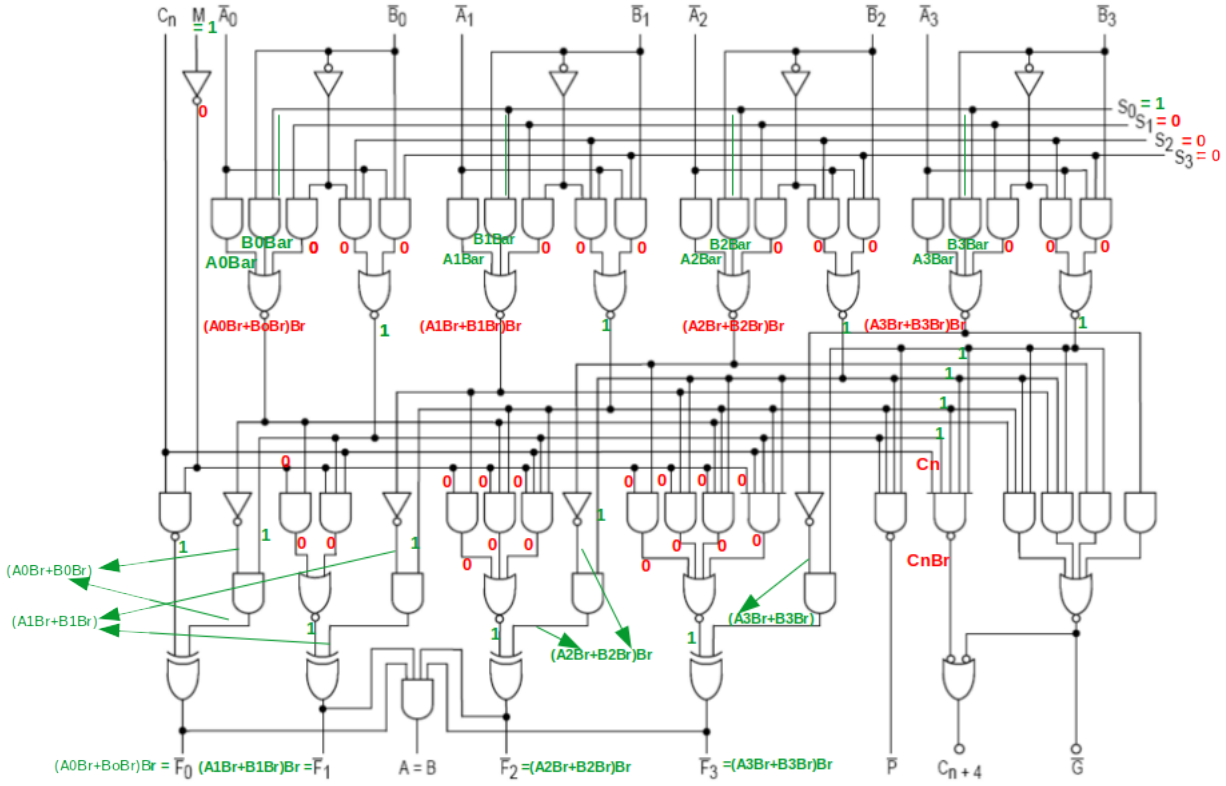
Solution:

In the picture above, the input is fed as inverted version. Note that the operation $F = \overline{(A+B)}$. If one considers the input as it is, then the output woud be as given below

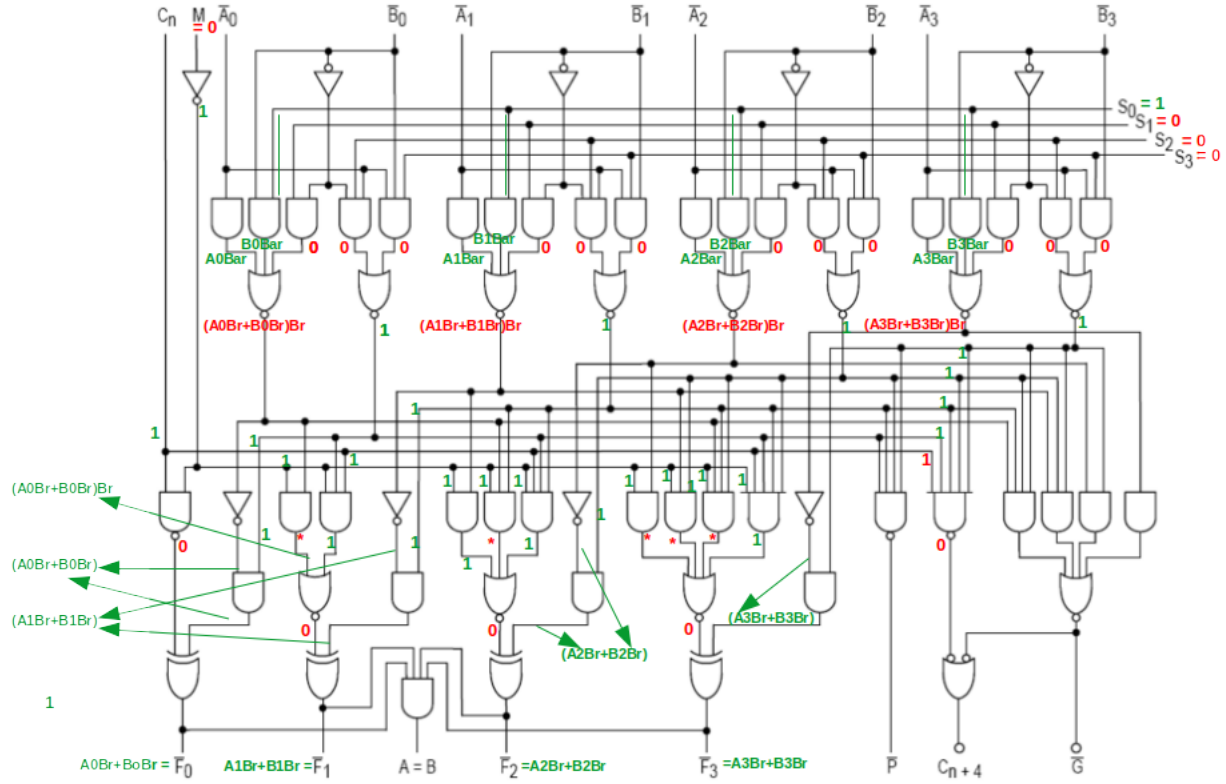
					Active-high logic	Active-high logic
	S_3	S_2	S_1	S_0	Logic $(M = 1)$	Arithmetic $(M = 0)(C_n = 1)$
	0	0	0	1	output $F = \overline{(A+B)}$	output $F = A + B$
	0	1	0	0	output $F = \overline{(AB)}$	output $F = A$ plus $A\overline{B}$
	0	1	1	0	output $F = A \bigoplus B$	output $F = A$ minus B
	1	0	0	1	output $F = \overline{A \bigoplus B}$	output $F = A$ plus B

The symbol '+' means logical OR. The words 'plus' & 'minus' are arithmetic '+' and '-'.

(a) The following Figure corresponds to logical operation $M = 1$



(b) The following figure corresponds to arithmetic operation $(M = 0)(C_n = 1)$



(c) For all other three entries, the evaluations could be done as above and could be verified.

9. Floating Point Unit (FPU):

   (a) Given the two positive floating point numbers (in IEEE 754 format), $N_1 = c_1 \times 10^{q_1}$ and $N_2 = c_2 \times 10^{q_2}$, give the table containing addition, subtraction, multiplication and division operations on these numbers.

   (b) Build an FPU unit, specify the internal circuitry of adder, subtractor, multiplier and specify the internal interconnection using various internal bus.

   Solution:

   See Section 9.7.1 for (a) and Section 9.7.3 for (b) all in Hamachar.